Table-top lab · guide

How a Sidewalk Is Made and How It Works

Cement is a fine gray powder that makes a paste with water, and it is only roughly 10 to 15 percent of the mix by volume. Concrete is cement plus sand, gravel and water. The cement and water react chemically (hydration), which is why concrete hardens even under water, and the slab is kept damp for about a week (curing) while most of its strength builds in the first month. Concrete shrinks slightly as it hardens, and grooves (control joints) give cracks a planned, neat line. Concrete is strong when squeezed and weak when bent, and a compacted gravel base helps support a slab evenly. Safety: wet concrete and dry cement powder can burn skin and irritate eyes and lungs, so children stay well back and never touch or taste either. The table-top activities use only flour, rice, chalk and crackers; check for food allergies, and keep the flour mix out of mouths. Chalk can snap into small pieces, so keep them off the floor.

Try 1 · What is a sidewalk made of?

Look closely at a sidewalk square outside, or at a photo of one. Find tiny stones, speckles and a groove. Draw one square on paper and mark where the groove runs.

Look for: A sidewalk is made of concrete, and concrete is a mix of four things: gravel, sand, cement and water. Cement is a fine gray powder. It is only a small part of the mix, roughly one part in eight, and when it meets water it makes a paste that coats the sand and stones.

Try 2 · How does wet concrete turn hard?

Stir two spoonfuls of flour with a little water in a paper cup to make a thick paste. Stir in a handful of dry rice so the paste coats every grain. Leave the cup for a day, then tap it and feel how hard it got. Flour paste only dries out, but real concrete hardens by a chemical reaction.

Look for: Wet concrete is poured into a wooden frame called a form, spread flat and smoothed. Then the cement and water begin a chemical reaction that links everything together. It hardens by that reaction, not just by drying out, which is why concrete can even harden under water.

Try 3 · Why is a sidewalk strong, and what holds it up?

Stand a stick of sidewalk chalk on end and press down on top with one finger to see that it holds. Now lay the chalk across two books and press gently in the middle until it snaps. Chalk is a stand-in for concrete: strong when squeezed, weak when bent.

Look for: Concrete is very strong when it is squeezed, like when feet and bike wheels press down on it. It is weaker when it bends or stretches. A layer of packed gravel under the slab helps support it evenly and spread the weight out, so the slab bends less.

Try 4 · Why does a sidewalk have grooves?

Look at the lines on a graham cracker. Break one cracker along a line. Then break another across the middle where there is no line. See which break is neat and straight.

Look for: A new slab shrinks a little as it hardens, and heat and cold make it grow and shrink all year. Concrete can crack when that happens. A groove makes the slab thinner along one line, so a crack tends to start there and run straight instead of wandering anywhere.

Print the lab sheet and this guide separately.